Quantum Field Theory II – Assignment 8

Exercise sheet 8, Advanced QFT, Prof. Jürgen Berges

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Problem 8.1 (Superficial degree of divergence for QED)

Review the Feynman rules of quantum electrodynamics and derive the superficial degree of divergence D for this theory. Proceed similarly to the derivation of D for the φ^4-theory as it was presented in the lecture.

The superficial degree of divergence D is given by the power of momenta in the highest order, i.e. most divergent, contribution to an amplitude. Every loop yields a momentum integral $\int \mathrm{d}^4 k_i$ and therefore $d = 4$ powers of momentum. All other powers derive from the propagators of internal lines. To specify these, we recall the QED Feynman rules for propagators

$$\text{internal fermion line:} \quad \xrightarrow{\ p\ } \hat{=} \frac{i(\not{p} + m_0)}{p^2 - m_0^2} = \frac{i(\not{p} + m_0)}{(\not{p} + m_0)(\not{p} - m_0)} = \frac{i}{\not{p} - m_0},$$

$$\text{internal photon line:} \quad \mu \sim\sim\sim\sim\sim\sim \nu \ (q) \ \hat{=} -\frac{i\eta_{\mu\nu}}{q^2},$$

$$\left(\text{vertex:} \quad \begin{array}{l} p_2 \searrow \\ \qquad \sim\sim\sim\ \mu \ (q) \\ p_1 \nearrow \end{array} \ \hat{=} -ie\gamma^\mu \delta^{(4)}(p_1 - p_2 - q) \right).$$

We expect D to be a function of any of the following

$$D = D\big(E_f, I_f, E_\gamma, I_\gamma, V, L\big),$$

where the arguments are the numbers of

$$\begin{aligned} &E_f : \text{external fermions}, && I_f : \text{internal fermions}, \\ &E_\gamma : \text{external photons}, && I_\gamma : \text{internal photons}, \\ &V : \text{QED vertices}, && L : \text{QED loops}. \end{aligned}$$

From the Feynman rules, we easily see the UV-structure of QED amplitudes is given by

$$A_{\text{UV}} \sim \int \frac{\prod_{l=1}^{L} \mathrm{d}^4 k_l}{\prod_{i=1}^{I_f} (\not{p}_i - m_0) \cdot \prod_{j=1}^{I_\gamma} q_j^2},$$

resulting in a superficial degree of divergence of

$$D = 4L - I_f - 2I_\gamma.$$

We proceed by inserting for the number of loop integrations

$$L = I_f + I_\gamma - (V - 1),$$

which again follows from the Feynman rules as they attribute one momentum integral to each internal line $I = I_f + I_\gamma$ while each vertex contributes one $\delta^{(4)}$, one of whom only enforces overall momentum integration. Thus

$$D = 4\big(I_f + I_\gamma - (V - 1)\big) - I_f - 2I_\gamma = 4 - 4V + 3I_f + 2I_\gamma.$$

The number of vertices can also be reexpressed by

$$V = 2I_\gamma + E_\gamma = \frac{1}{2}\left(2I_f + E_f\right),$$

since every QED vertex has one photon- and two fermion-lines, and every internal line is connected to two vertices. Altogether, we get

$$\begin{aligned} D &= 4 - \left(2I_\gamma + E_\gamma\right) - 3 \cdot \frac{1}{2}\left(2I_f + E_f\right) + 3I_f + 2I_\gamma \\ &= 4 - E_\gamma - \frac{3}{2}E_f. \end{aligned}$$

Additional note: Looking at $\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \overline{\psi}(i\gamma\cdot\partial - m_0)\psi - eA_\mu\overline{\psi}\gamma^\mu\psi$, we see that since $[\psi] = \frac{3}{2}$, $[A_\mu] = 1$, QED's coupling has mass-dimension $[e] = 0$. From this, we conclude that QED is renormalizable in 4, i.e. has finitely many divergencies, but at every order in perturbation theory.

Problem 8.2 (One-loop structure of QED)

From problem 8.1, you should have found that the superficial degree of divergence of a QED-amplitude with an interaction-Lagrangian $\mathcal{L}_{\text{int}} = \overline{\psi}\gamma^\mu A_\mu\psi$ is

$$D = 4 - E_\gamma - \frac{3}{2}E_f. \tag{1}$$

Since $[e] = 0$, the theory is renormalizable.

a) Give D for each of the following amplitudes.

i) The vacuum energy.



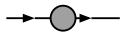
$E_f = E_\gamma = 0, \qquad D = 4 - E_\gamma - \frac{3}{2}E_f = 4$

ii) The photon propagator.



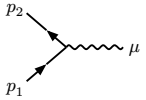
$E_f = 0, E_\gamma = 2, \quad D = 4 - E_\gamma = 2$

iii) The electron propagator.



$E_f = 2, E_\gamma = 0, \quad D = 4 - \frac{3}{2}E_f = 1$

iv) The electron-electron-photon vertex.



$E_f = 2, E_\gamma = 1, \quad D = 4 - E_\gamma - \frac{3}{2}E_f = 0$

v) The 1-photon amplitude.



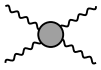
$E_f = 0, E_\gamma = 1, \quad D = 4 - E_\gamma = 3$

vi) The 3-photon amplitude.



$$E_f = 0, E_\gamma = 3, \quad D = 4 - E_\gamma = 1$$

vii) The 4-photon amplitude.



$$E_f = 0, E_\gamma = 4, \quad D = 4 - E_\gamma = 0$$

b) Verify explicitly that the one-loop diagram contributing to the one-point function vanishes using the discrete symmetry of charge conjugation, $j^\mu \to -j^\mu$, $A^\mu \to -A^\mu$. There are two Feynman diagrams contributing to the three-point function at one-loop order. Show that these cancel. Show that the diagrams contributing to any n-point photon amplitude, for n odd, cancel in pairs.

Charge conjugation $\hat{C}$ is a symmetry of QED, meaning

$$\hat{C}\ |\Omega\rangle = |\Omega\rangle.$$

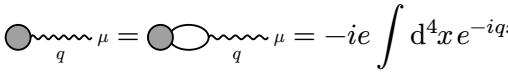
Since $j^\mu(x)$ changes sign under charge conjugation,

$$\hat{C} j^\mu(x) \hat{C}^\dagger = -j^\mu(x), \quad \text{where } \hat{C}^\dagger \hat{C} = \mathrm{id},$$

we infer that its vacuum expectation value vanishes at one-loop (and any other number of loops),

$$\begin{aligned}\langle \Omega \mid T j^\mu(x) \mid \Omega \rangle = \langle \Omega \mid j^\mu(x) \mid \Omega \rangle = \langle \Omega \mid \hat{C}^\dagger \hat{C} j^\mu(x) \hat{C}^\dagger \hat{C} \mid \Omega \rangle \\ = -\langle \Omega \mid j^\mu(x) \mid \Omega \rangle = 0.\end{aligned}$$

This carries over to the photon one-point function



$$= -ie \int \mathrm{d}^4x\, e^{-iqx} \langle \Omega \mid T j^\mu(x) \mid \Omega \rangle = 0,$$

where $j^\mu(x) = \overline{\psi}(x)\gamma^\mu\psi(x)$.

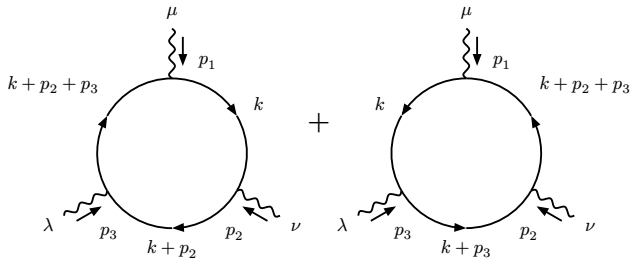
More explicitly, we may also write the one-point amplitude at one-loop as

$$\Gamma^{(1)\mu}|_{\text{1-loop}} = -ie \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{-i\ \mathrm{tr}[\gamma^\mu(\not{k} + m)]}{k^2 - m^2} = -e \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{4k^\mu}{k^2 - m^2} = 0,$$

where we used $\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ and $\mathrm{tr}(\gamma^\mu) = 0$ (see pr. 9.3, QFT I), and which vanishes, since k^μ is odd and $\frac{1}{k^2-m^2}$ is even.

The three-point amplitude at one-loop consists of the two diagrams

$$i\Gamma^{(3)\mu\nu\lambda}|_{\text{1-loop}} =$$



$$= -(-ie)^3 \int \frac{\mathrm{d}^4k}{(2\pi)^4} \left\{ \mathrm{tr}[\gamma^\mu S_F(k) \gamma^\nu S_F(k + p_2) \gamma^\lambda S_F(k + p_2 + p_3)] \right.$$

$$+ \operatorname{tr}[\gamma^\mu S_F(-k)\gamma^\nu S_F(-k-p_2)\gamma^\lambda S_F(-k-p_2-p_3)]\},$$

where $S_F(k) \coloneqq \frac{1}{\not{k}-m} = \frac{\not{k}+m}{k^2-m^2}$. To show that this vanishes, we use γ^μ's and S_F's behavior under charge conjugation,

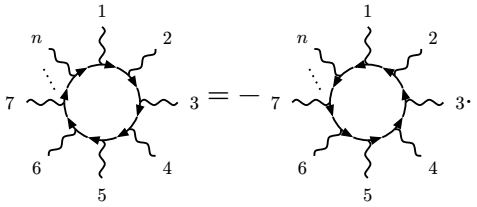
$$\hat{C}^\dagger \gamma^\mu \hat{C} = (\gamma^\mu)^T, \quad \hat{C}^\dagger S_F(k)\hat{C} = -S_F(-k),$$

and the cyclicity of the trace (to send the last $\hat{C}^\dagger$ to the front). Thus

$$\begin{aligned}
&\operatorname{tr}[\gamma^\mu \underbrace{\hat{C}\hat{C}^\dagger}_{\text{id}} S_F(-k)\hat{C}\hat{C}^\dagger\gamma^\nu\hat{C}\hat{C}^\dagger S_F(-k-p_2)\hat{C}\hat{C}^\dagger\gamma^\lambda\hat{C}\hat{C}^\dagger S_F(-k-p_2-p_3)\hat{C}\hat{C}^\dagger] \\
&= \operatorname{tr}\Big[(\gamma^\mu)^T(-S_F(k))(\gamma^\nu)^T(-S_F(k+p_2))\big(\gamma^\lambda\big)^T(-S_F(k+p_2+p_3))\Big] \\
&= -\operatorname{tr}[\gamma^\mu S_F(k)\gamma^\nu S_F(k+p_2)\gamma^\lambda S_F(k+p_2+p_3)].
\end{aligned}$$

Reinserting into $i\Gamma^{(3)}|_{\text{1-loop}}$ yields immediately $i\Gamma^{(3)}|_{\text{1-loop}} = 0$.

We turn to the general case of all diagrams contributing to a photon n-point function where n is odd. First note, that every fermionic loop with an odd number of vertices has a counterpart in which the momenta point along the loop in opposite direction

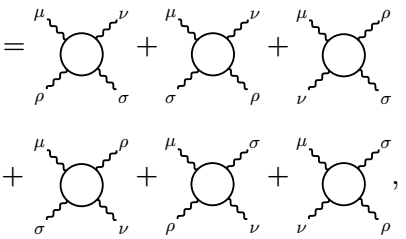


Since an n-point photon function receives contributions from $(n-1)!$, and $n!$ is even for $n > 2$ (the case $n = 1$, i.e. the one-loop diagram , we already showed to be vanishing in the beginning)

we can always arrange all contributing diagrams in pairs of two, where for the special case of n odd, we may partner each diagram with its oppositely running counterpart in order to let each pair of our partition vanish individually.

c) The photon four-point amplitude is a sum of six diagrams. Show explicitly that the potential logarithmic divergences of these diagrams cancel.

The six diagrams are

$\Gamma^{(4)\mu\nu\rho\sigma} =$ 

where we omit momentum arrows in the fermion loops since clock- and counterclockwise running are equal for an even number of vertices.

The potential log-divergence of $i\Gamma^{(4)\mu\nu\rho\sigma}$ stems entirely from integrating over the internal momentum, i.e.

$$i\Gamma^{(4)\mu\nu\rho\sigma}_{\text{div}} \sim \int \mathrm{d}^4k \, \frac{(\not{k})^4}{(k^2)^4},$$

and is the same irrespective of a change in external momenta p_i, $i \in \{1,2,3,4\}$, or, for that matter, the fermion mass m. To speed up calculations, we may therefore set both to zero.

The divergent part $i\Gamma^{(4)\mu\nu\rho\sigma}_{\text{div}}$ of the photon 4-point amplitude may then be written much more compactly as

$$\begin{aligned} i\Gamma^{(4)\mu\nu\rho\sigma}_{\text{div}} = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{-1}{(k^2)^4} [&\operatorname{tr}(\gamma^\mu \not{k} \gamma^\nu \not{k} \gamma^\rho \not{k} \gamma^\sigma \not{k}) + \operatorname{tr}(\gamma^\mu \not{k} \gamma^\nu \not{k} \gamma^\sigma \not{k} \gamma^\rho \not{k}) \\ &+ \operatorname{tr}(\gamma^\mu \not{k} \gamma^\rho \not{k} \gamma^\nu \not{k} \gamma^\sigma \not{k}) + \operatorname{tr}(\gamma^\mu \not{k} \gamma^\rho \not{k} \gamma^\sigma \not{k} \gamma^\nu \not{k}) \\ &+ \operatorname{tr}(\gamma^\mu \not{k} \gamma^\sigma \not{k} \gamma^\nu \not{k} \gamma^\rho \not{k}) + \operatorname{tr}(\gamma^\mu \not{k} \gamma^\sigma \not{k} \gamma^\rho \not{k} \gamma^\nu \not{k})]. \end{aligned}$$

To get a grasp on this expression, we concern ourselves solely with the first trace initially. Using the gamma-matrices' anti-commutation relations and some of the trace identities worked out in problem 9.3 of QFT I, it can be expressed as

$$\begin{aligned} \operatorname{tr}(\gamma^\mu \not{k} \gamma^\nu \not{k} \gamma^\rho \not{k} \gamma^\sigma \not{k}) = 32k^\mu k^\nu k^\rho k^\sigma - 8k^2(&k^\mu k^\nu g^{\rho\sigma} + k^\rho k^\sigma g^{\mu\nu} \\ &+ k^\mu k^\sigma g^{\nu\rho} + k^\nu k^\rho g^{\mu\sigma}) + 4k^4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}). \end{aligned}$$

We symmetrize products of k by sending $k^\mu k^\nu \to \frac{k^2}{4} g^{\mu\nu}$ and $k^\mu k^\nu k^\rho k^\sigma \to \frac{k^4}{24}(g^{\mu\nu}g^{\rho\sigma} + g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$ to obtain

$$\operatorname{tr}(\gamma^\mu \not{k} \gamma^\nu \not{k} \gamma^\rho \not{k} \gamma^\sigma \not{k}) = \frac{4}{3} k^4 (g^{\mu\nu}g^{\rho\sigma} - 2g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}).$$

From this expression, the other five traces follow simply by permutations of indices. Since the metric is symmetric, they all sum up to zero.